

# Neural Population Dynamics Reveal Internal Representations Underlying Adaptive Obstacle Avoidance in Reinforcement Learning

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## Abstract

When a controller must steer toward a goal while unexpected forces deflect its path, what determines whether it recovers or fails? This question sits at the intersection of motor neuroscience and autonomous control, and it remains difficult to answer from behavioural data alone because the internal signals that drive moment-to-moment corrections are not directly observable. We developed a computational framework that places a reinforcement learning controller – trained through the same kind of trial-and-error refinement that shapes biological motor learning – in a reaching-inspired obstacle-avoidance task and perturbs it with graded lateral force impulses, mirroring protocols used to study feedback control in human arm movements. Rather than reporting only task success, we treat the controller’s hidden-layer activity as a neural population recording and apply dimensionality reduction, state-space trajectory analysis, and linear decoding to ask what internal representations track behaviour and distinguish recovery from failure.

We find that the controller adapts successfully to moderate perturbations across all directions but fails progressively under strong rightward disturbances, revealing a direction-dependent asymmetry in robustness. The hidden-layer activity organises into a compact low-dimensional manifold – accounting for the majority of population variance in just two dimensions – and subdivides into discrete clusters that align with phases of the movement rather than with specific perturbation conditions. Lateral velocity is clearly encoded across the population, and the most extreme perturbation conditions produce latent states that a simple linear classifier can separate from one another, whereas mild conditions remain geometrically indistinct. These properties closely mirror those documented in motor cortex during biological reaching: compact manifold geometry, phase-structured clustering, and distributed encoding of kinematic variables. Our results suggest that reward-driven optimisation spontaneously generates representational organisation similar to biological motor populations, and that population-style analyses of artificial controllers can reveal not just whether adaptive behaviour succeeds but why it sometimes does not.

## 1 Introduction

The ability to reach toward a target while correcting for unexpected disturbances is a hallmark of biological motor control [1, 2, 3, 4]. When the arm is pushed mid-flight, the nervous system does not wait for a deliberate decision; corrections emerge within tens of milliseconds, shaped by the continuous interplay between sensory feedback and internal predictions [5, 6, 7]. These fast, flexible responses arise from coordinated activity across distributed neural populations in motor cortex and cerebellum rather than from single-neuron specialisation [8, 9, 10, 11]. Understanding how population-level organisation gives rise to robust adaptive behaviour is a

central goal of systems motor neuroscience.

Reinforcement learning (RL) offers a complementary perspective. In RL, a controller is not designed with explicit rules; instead, it discovers its own movement strategy through repeated interaction with the environment, receiving a reward signal whenever it performs well and adjusting accordingly [12, 13]. This is, in spirit, how biological motor skill is acquired: practice generates outcome feedback, and the nervous system gradually updates its internal policy to maximise success [14, 15].

**What is PPO, and why does it matter here?** The specific algorithm we use is Proximal Policy Optimization (PPO) [16]. For readers more at home in a neuroscience lab than a machine learning classroom, the clearest analogy is incremental motor learning. Imagine a participant practising a novel reaching task: each trial, the motor system asks whether the movement was better or worse than usual, and nudges the control policy slightly toward strategies that worked. Experienced clinicians and motor learning researchers know that overcorrecting after a single poor trial is counterproductive – it destabilises what already works. PPO builds this conservatism into its update rule. After each batch of training trials, it computes how much better or worse the new strategy would have performed relative to the old one, then limits the size of the update so the policy never changes too drastically in a single step. This is the meaning of “proximal”: the new policy stays close to the current one, much as incremental motor learning keeps refined movements anchored to the existing motor programme. The result is a learner that improves steadily without the wild swings in performance that unconstrained gradient updates can produce.

Most RL studies evaluate controllers solely on task outcomes – goal success rate, cumulative reward – while treating the network as a black box [17, 18]. This is informative but incomplete: knowing that a controller succeeds nine times out of ten tells you nothing about what it does differently on the tenth, or which internal signals predict whether a given perturbation will be absorbed or cause failure. The same interpretability problem motivates population analyses in biological motor neuroscience: task performance measures confirm that a subject compensates successfully, but the population dynamics reveal *how* and *why* [8, 19, 10].

Recent work has shown that motor cortex activity during reaching unfolds on low-dimensional neural manifolds – smooth subspaces within the high-dimensional activity space that capture the essential dynamics of movement generation [8, 9, 19, 10]. These manifolds can be identified by principal component analysis (PCA), their structure interrogated with trajectory analyses, and behaviourally relevant variables decoded from them with linear classifiers [19, 14, 20, 21]. Whether an artificially trained controller develops similar organisation is an open question with practical implications for interpretability [22, 23].

Here, we train a PPO controller on an obstacle-avoidance reaching task and probe it with the same graded perturbation protocol used in biological reaching experiments. We record hidden-layer activity during evaluation and analyse it using dimensionality reduction, trajectory visualisation, and linear decoding. Our goal is to identify which representational features emerge from reward-driven training, whether they parallel those in motor cortex, and which features distinguish successful recovery from failure.

Our contributions are: (i) a population-analysis workflow applied to a trained RL controller enabling direct comparison with biological motor data; (ii) evidence of compact low-dimensional manifold structure consistent with neural manifold dynamics in motor cortex; (iii) identification of a direction-dependent asymmetry in robustness visible in both behaviour and latent geometry; and (iv) demonstration that extreme perturbation conditions produce linearly separable latent representations while mild conditions do not, mirroring difficulty-dependent separability in biological populations.

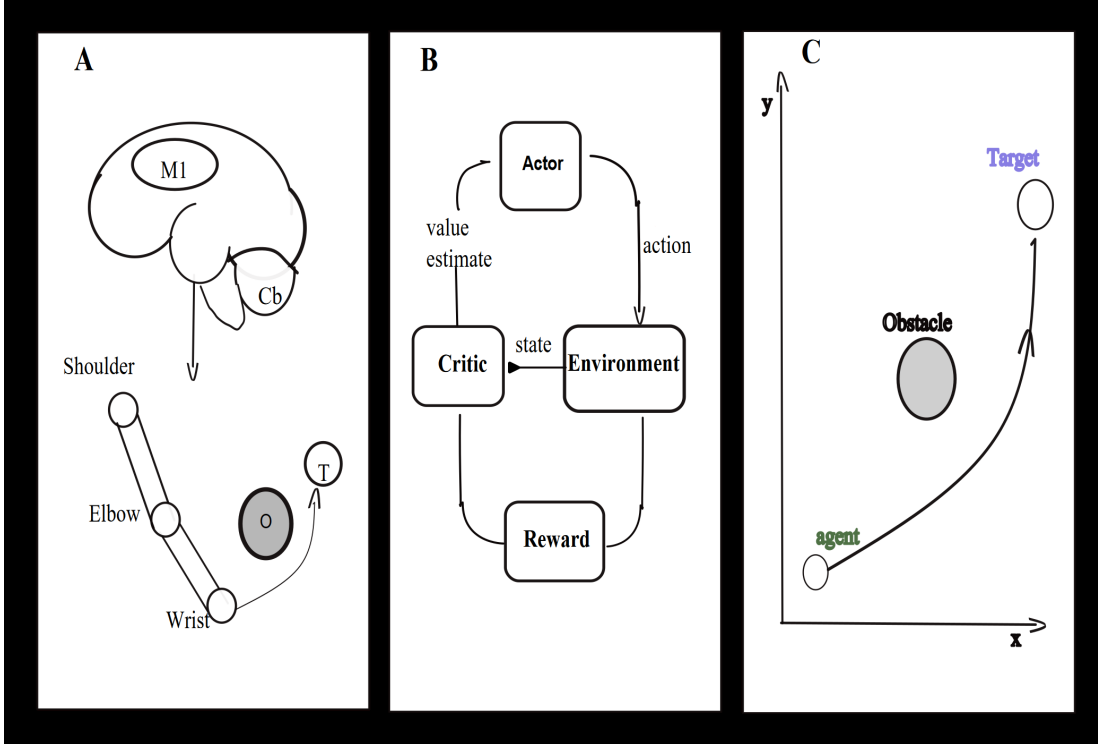


Figure 1: **Task, control architecture, and analysis pipeline.** (A) Biological analogy: motor cortex (M1) and cerebellum (Cb) drive a multi-joint arm toward a target (T) while avoiding an obstacle (O). This panel is used for conceptual grounding; all experiments use a point-mass agent. (B) The PPO actor-critic loop: the actor selects a continuous force command; the environment advances the agent and returns the next state; the critic estimates the long-term value of that state; a scalar reward drives both networks. (C) Task geometry: the agent navigates from a fixed start through a two-obstacle corridor to a fixed goal on a 2-D plane.

## 2 Methodology

### 2.1 Task Environment and State Representation

We implemented the task using the Gymnasium interface [24]. The agent is a two-dimensional point mass controlled by continuous force commands, evolving under Newtonian dynamics. Dynamics were kept deliberately minimal so that any representational structure found in the network reflects learning rather than physical constraints. Figure 1 shows the task schematically; Panel A uses a planar arm analogy for biological grounding, and all experiments use the point-mass agent in Panel C.

The main experiment used a corridor formed by two static circular obstacles. A preliminary single-obstacle variant established the basic obstacle-avoidance skill before the two-obstacle and perturbation protocol were introduced. The agent receives an agent-centred observation at each timestep:

$$s_t = [g_x - x_t, g_y - y_t, o_{Lx} - x_t, o_{Ly} - y_t, o_{Rx} - x_t, o_{Ry} - y_t, v_{x,t}, v_{y,t}] \quad (1)$$

where  $(x_t, y_t)$  is the agent’s position,  $(v_{x,t}, v_{y,t})$  its velocity,  $(g_x, g_y)$  the goal, and  $(o_L, o_R)$  the obstacle centres.

## 2.2 Reward Structure and Perturbation Protocol

The reward at each timestep balances four objectives:

$$R_t = w_p \Delta d_{\text{goal}} - w_v \|v_t\| - w_c C_t + w_s S_t \quad (2)$$

Progress toward the goal is rewarded; excessive speed penalised; collision incurs a large negative penalty; reaching the goal earns a terminal bonus.

Lateral force impulses were applied mid-movement for six timesteps:

$$f_x(t) = p_i, \quad t \in [50, 55], \quad p_i \in \{\pm 0.15, \pm 0.30, \pm 0.45\} \quad (3)$$

defining seven conditions: control (P0) and three increasing leftward (L1–L3) and three rightward (R1–R3) impulses. The design mirrors perturbation protocols used in human reaching studies to isolate online feedback correction [1, 2, 3, 5].

## 2.3 Reinforcement Learning Controller

All training used Stable-Baselines3 [25] with PPO [16]. PPO maximises a clipped surrogate objective:

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[ \min(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t) \right] \quad (4)$$

where  $r_t(\theta)$  is the probability ratio of updated to old policy,  $\hat{A}_t$  the estimated advantage, and  $\epsilon = 0.2$  the clipping threshold.

The controller uses separate policy and value networks, each a feedforward MLP with two hidden layers of 256 units. Continuous actions are drawn from a diagonal Gaussian; observations were normalised online (VecNormalize). Training ran for  $2 \times 10^6$  timesteps with learning rate  $3 \times 10^{-4}$ , discount  $\gamma = 0.99$ , entropy coefficient 0.005, value-function coefficient 0.5, and gradient norm clipped at 0.5. All experiments used Stable-Baselines3 v2.8.0, Gymnasium, and PyTorch on an Intel® Core™ Ultra 7 255H processor.

## 2.4 Neural Population Analysis

Following training, the activation vector  $h_t \in \mathbb{R}^{256}$  of the first hidden layer was extracted at every evaluation timestep, treating the collection  $H = \{h_1, \dots, h_T\}$  as a population recording. Three analyses were applied: **PCA** revealed the geometry of the population state space; **tuning analysis** related unit activity to instantaneous lateral velocity; and **linear decoding** tested whether perturbation condition is linearly readable from single-timestep hidden activity. All analyses used Python (NumPy, SciPy, scikit-learn).

# 3 Results

We report four sets of findings: behavioural adaptation and its limits; kinematic signatures; latent state-space structure; and linear decodability.

## 3.1 Behavioural Adaptation and Its Limits

Figure 2 overlays reach trajectories across all seven conditions. Under the unperturbed control and all mild perturbations the controller reached the goal on every trial, generating smooth paths that curved

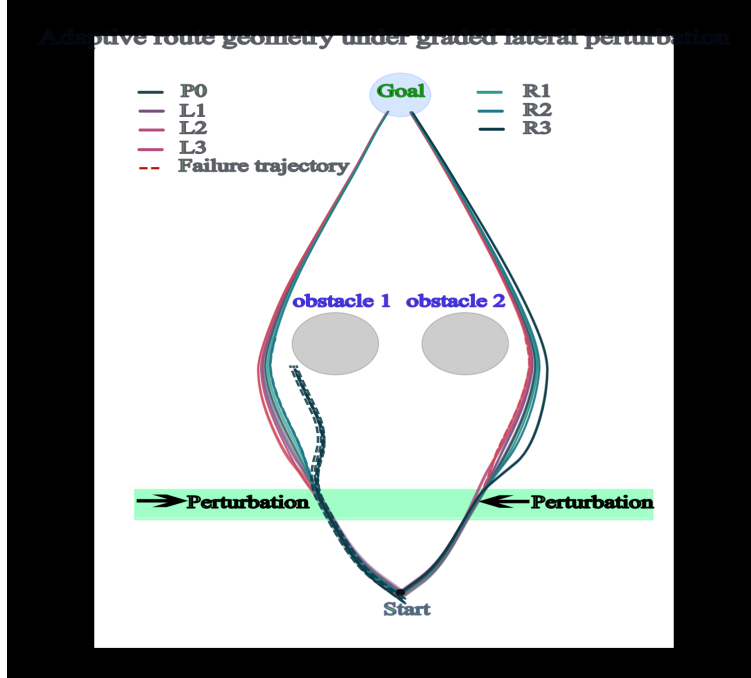


Figure 2: **Adaptive route geometry under graded lateral perturbation.** All evaluation episodes are overlaid per condition. Solid lines are successful trials; the dashed red line is a failure trajectory. Leftward conditions (L1–L3, mauve/pink) shift paths left; rightward conditions (R1–R3, teal/dark green) shift them right. The shaded green band marks the workspace segment where the force impulse is applied; grey ellipses are the two static obstacles. Goal success was maintained across control and all mild conditions, declining only under the two strongest rightward disturbances and the strongest leftward impulse, with the rightward direction proving disproportionately disruptive.

symmetrically around both obstacles. Trajectories deflected proportionally in the direction of the applied force, bowing left under leftward conditions and right under rightward ones.

Failures occurred exclusively under the two strongest rightward conditions and the largest leftward impulse, with rightward disturbances proving more disruptive than leftward ones of equal magnitude. This asymmetry is a geometric consequence of the learned trajectory: it runs slightly to the left of corridor centre, giving more margin to absorb leftward pushes before approaching an obstacle boundary. Across all successful trials, peak speed was effectively invariant across conditions – adaptation was achieved through path redirection, not acceleration.

### 3.2 Kinematic Signatures of Perturbation Direction

Lateral velocity (Figure 3A) separated cleanly by direction throughout the perturbation window and scaled with magnitude. Conditions on the same side of zero partially reconverged afterwards, indicating a proportional rather than binary corrective response. Total speed (Figure 3B) was indistinguishable across conditions – the controller preserves movement tempo while redirecting the path, a kinematic signature consistent with path-correction strategies in human online reaching [1, 2, 26, 4].

### 3.3 Low-Dimensional Structure of the Latent State Space

Two findings stand out from the PCA (Figure 4). First, 74.8% of the variance in a 256-unit network is captured by two dimensions alone, indicating a compact low-dimensional representation consistent with neural manifold

structure in motor cortex [8, 9, 19, 10]. Second, the manifold subdivides into four spatially coherent clusters that do not align with individual perturbation conditions; they likely reflect discrete movement phases shared across all trials. Perturbation-specific differences appear as density shifts within this shared geometry rather than as separate manifolds, suggesting a common computational substrate for obstacle avoidance that is modulated by perturbation context.

### 3.4 Linear Decoding of Perturbation Context

A linear classifier trained on single-timestep hidden activity (Figure 5) decoded the most extreme perturbation conditions most accurately: the largest leftward impulse was correctly identified in roughly seven out of ten timesteps, and the largest rightward impulse in roughly six out of ten. Moderate rightward conditions were also decoded above chance. The unperturbed control and mild leftward conditions were most often misclassified as the largest leftward condition, reflecting their geometric proximity in the manifold. The success of linear decoding at all indicates that perturbation context is encoded in a geometrically structured, directly accessible form – not buried in nonlinear interactions among units – mirroring the linear decodability of task context from motor cortex population activity during biological reaching [8, 2, 14].

### 3.5 Learning Trajectory and Stability

Reward rose rapidly within the first third of the training budget and then plateaued stably (Figure 6). Evaluation reward across independently sampled checkpoints was tightly distributed around the plateau value, confirming that the representational analyses in Sections 3.3 and 3.4 reflect a well-converged rather than partially-trained controller.

## 4 Discussion

This study asked whether a PPO controller trained for adaptive obstacle avoidance develops internal representations that are interpretable in the same terms used to study biological motor populations. Across behaviour, kinematics, latent geometry, and linear decodability, the findings are consistently affirmative – with qualifications that deserve careful statement.

**Behavioural adaptation mirrors biological feedback correction.** The controller maintained robust performance under mild perturbations and degraded progressively under stronger ones, with a direction-dependent asymmetry attributable to the geometry of the learned corridor trajectory. Critically, the movement tempo – measured by total speed – was preserved across all conditions; adaptation appeared entirely in path geometry. This separation between corrective redirection and preserved speed is the same kinematic signature documented for online feedback corrections in human reaching, where motor cortex updates movement direction without imposing systematic speed changes [1, 2, 26, 4, 3]. We do not claim that the controller mimics a biological motor system; its dynamics are far simpler than those of a real limb, and it was optimised for reward rather than biological fidelity. Nevertheless, the convergence of the kinematic signature is consistent with optimal feedback control principles [5, 6] and suggests that this particular strategy may be a computationally preferred solution whenever a system must redirect a moving body under time pressure.

**Low-dimensional latent structure and its biological parallel.** The finding that the majority of variance in a 256-unit network collapses onto two dimensions is notable because the network was never constrained to be low-dimensional. In biological motor cortex, the dominant computational subspace during reaching spans far fewer dimensions than the number of recorded neurons, a property attributed to the structured, repetitive nature of movement [8, 9, 19, 10, 20, 21]. Our results suggest that a similar pressure operates in

reward-driven learning: because successful reaches share a common temporal structure, the network represents the essential state of the movement compactly, with perturbation context appearing as a modulation of shared geometry. The four-cluster decomposition reinforces this: the clusters align with movement phases rather than conditions, consistent with phase-structured dynamics in motor cortex [8, 9]. We note that the  $K = 4$  solution was identified post hoc; replication across multiple training seeds and held-out data would be needed to confirm whether this is a stable feature of the learned representation.

**Decoding and what it implies about internal organisation.** The pattern of decoder performance – reliable for extreme conditions, near-chance for mild ones – is not a simple decoding failure. It is a property of the manifold geometry: mild perturbations generate hidden states that are geometrically indistinguishable from larger perturbations that were also well corrected, because the controller handled them with similar small adjustments. This has a direct parallel in biological motor cortex, where the decodability of perturbation magnitude improves when classifiers are given a temporal window of population activity rather than a single snapshot [2, 8]. The practical implication for both artificial and biological systems is that a single snapshot of population activity is not always sufficient to identify the perturbation magnitude; understanding the source of failure requires following the representational trajectory over time.

**Limitations.** Results are from a single training seed; variability across runs is unknown. The environment is simplified: two-dimensional, static obstacles, a fixed perturbation window, and point-mass dynamics far simpler than a real limb. The feedforward MLP has no recurrence, limiting its ability to maintain a running internal estimate of perturbation history. PCA and decoding are descriptive; they confirm that structure exists but do not establish causal links between specific units and behaviour. Large language models were used to support editing, proofreading, and L<sup>A</sup>T<sub>E</sub>X formatting; they were not used to generate data, run analyses, or draw scientific conclusions.

**Implications for biological and artificial systems.** Our results contribute to growing evidence that the organisational principles of biological motor populations – compact manifold geometry, phase-structured latent dynamics, and linear decodability of task context – are not unique to neural circuits shaped by evolution and development. They can emerge spontaneously from reward-driven optimisation in artificial networks [8, 10, 14, 11, 20]. This convergence is meaningful in two directions. For neuroscientists, it validates using neuroscience-inspired analytical tools on artificial controllers, enabling the same interpretive language applied to motor cortex to be used for autonomous systems. For engineers and clinicians working on assistive or rehabilitation technologies – where understanding the relationship between internal representations, behavioural adaptation, and movement recovery is practically important – it suggests that these analytical tools may provide actionable diagnostic information beyond simple task-success metrics. Understanding where and how representational structure breaks down under strong perturbations could inform both our understanding of motor cortex failure modes and the design of more robust, interpretable controllers.

## 5 Conclusion

We trained a PPO controller on an obstacle-avoidance reaching task and analysed its internal representations using the population-level toolkit of motor neuroscience. The controller adapted to moderate perturbations through path-level corrections that preserved movement speed, with robustness declining asymmetrically under strong rightward disturbances. The hidden-layer activity organised into a compact low-dimensional manifold – capturing the large majority of population variance in two dimensions – with four phase-aligned clusters shared across conditions. Extreme perturbation conditions produced latent states separable by a

linear classifier; mild conditions did not, a pattern mirroring difficulty-dependent decodability in biological motor cortex.

These findings demonstrate that reward-driven learning generates structured internal representations with organisational properties consistent with biological motor populations. Treating policy hidden activity as a population signal extends behavioural evaluation and may help explain why adaptive controllers succeed or fail under conditions they were not explicitly trained on. Future work will extend this framework to recurrent architectures, richer environments, and multiple training seeds, and will test whether the representational structure identified here predicts behavioural failure across task variants.

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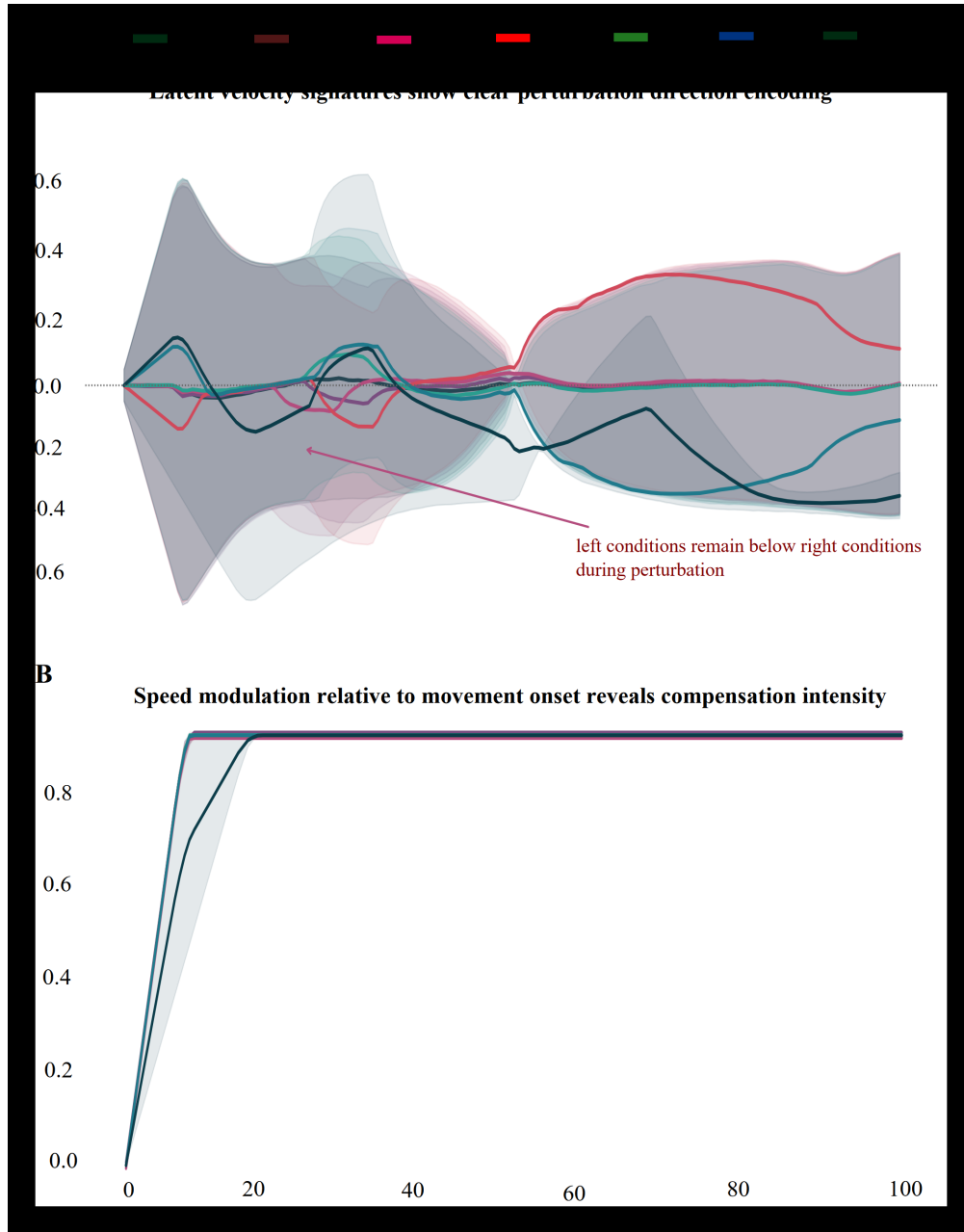


Figure 3: **Velocity profiles across perturbation conditions.** **(A)** Mean lateral velocity over time (shaded:  $\pm 1$  SD). Leftward conditions drive negative lateral velocity; rightward drive positive. Conditions diverge clearly during the perturbation window (timesteps 50–55) and partially reconverge thereafter, consistent with a proportional corrective response. **(B)** Total speed as a function of normalised movement phase. Speed rises rapidly at onset and plateaus consistently across all seven conditions, confirming that perturbation correction is expressed through lateral redirection rather than speed change.

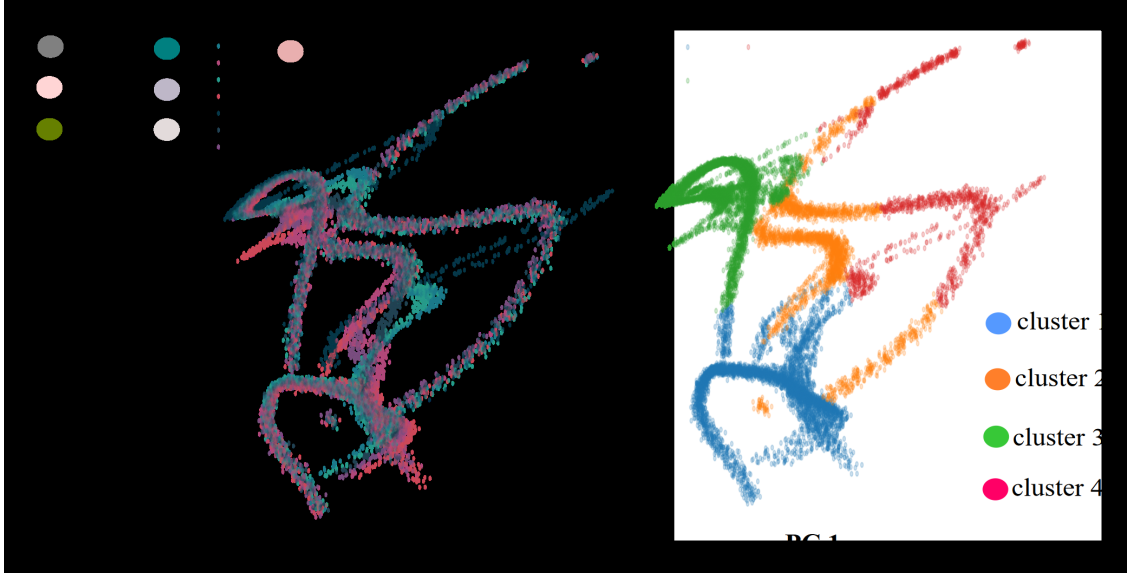


Figure 4: **Low-dimensional structure of the policy hidden-layer activity.** Each point is the 256-dimensional hidden-state at one timestep projected onto the first two principal components. **(A)** Coloured by perturbation condition. The manifold has a consistent curved geometry; extreme conditions occupy partially separable peripheral regions.  $PC1 = 46.8\%$ ,  $PC2 = 28.0\%$ , total =  $74.8\%$ . **(B)** Coloured by  $K$ -means cluster label ( $K = 4$ ). The manifold divides into four spatially coherent regions likely corresponding to movement phases – approach, perturbation, correction, arrival – shared across all conditions.

|   |      |      |      |      |      |      |      |
|---|------|------|------|------|------|------|------|
| 0 | 0.00 | 0.10 | 0.03 | 0.20 | 0.51 | 0.00 | 0.15 |
| 1 | 0.00 | 0.12 | 0.07 | 0.35 | 0.33 | 0.00 | 0.12 |
| 2 | 0.01 | 0.08 | 0.11 | 0.43 | 0.30 | 0.00 | 0.07 |
| 3 | 0.00 | 0.04 | 0.04 | 0.70 | 0.17 | 0.00 | 0.05 |
| 4 | 0.01 | 0.03 | 0.02 | 0.22 | 0.53 | 0.01 | 0.18 |
| 5 | 0.01 | 0.04 | 0.01 | 0.20 | 0.50 | 0.04 | 0.20 |
| 6 | 0.01 | 0.03 | 0.02 | 0.18 | 0.13 | 0.01 | 0.62 |

Figure 5: **Linear decoding of perturbation condition from hidden-layer activity.** Rows are the true condition; columns are the classifier’s prediction. Diagonal entries (darker blue) are correct classifications. The largest leftward (L3) and rightward (R3) conditions are decoded most reliably. The unperturbed control and mild leftward conditions are frequently predicted as L3, reflecting their geometric proximity to the L3 cluster in the latent manifold rather than a simple decoding failure.

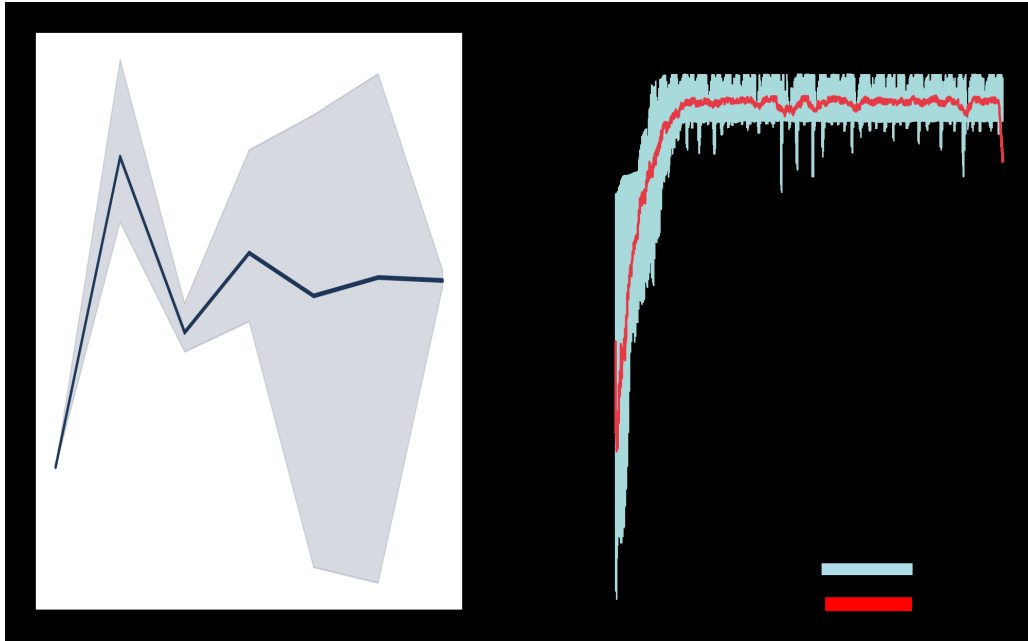


Figure 6: **Learning stability and policy improvement.** **(Left)** Evaluation reward across training checkpoints (mean  $\pm$  SD) showing stable performance after convergence. **(Right)** Episode reward over the full training run: raw per-episode values (light blue) and a smoothed trace (red) confirm rapid improvement in the first third of training and a stable plateau thereafter. The representations analysed in Figures 4–5 were extracted from this converged policy.